

STATEMENT OF WORK (SERVICES) (SOW)

GENERAL INFORMATION

Title of Project:

Preparation of chemical compounds needed for a Janus kinase 2 (JAK2) enzyme project

Statement of Need and Purpose:

The compounds described below are needed for evaluation in brain cancer models in assay models at NCATS. They will allow us to determine the compound targets and provide insight into the specificity for brain cancer cells.

Background Information and Objective:

JAK2 inhibitors are medications that block the activity of the Janus kinase 2 (JAK2) enzyme, which is involved in signaling pathways related to inflammation and immune response. These inhibitors are used to treat various conditions, including some cancers and inflammatory diseases like rheumatoid arthritis and atopic dermatitis. We have discovered a compound that targets brain cancer cells and we require the synthesis of it an analog for evaluation in various brain-cancer cell targeted assays. These two compounds will be evaluated for their potential for treatment in brain cancer.

- (1) 4-amino-N,N-dicyclopropyl-6-ethyl-1-methyl-1,6-dihydroimidazo[4,5-d]pyrrolo[2,3-b]pyridine-7-carboxamide; and
- (2) N,N-dicyclopropyl-4-((1,5-dimethyl-1H-pyrazol-3-yl)amino)-6-ethyl-1-methyl-1,6-dihydroimidazo[4,5-d]pyrrolo[2,3-b]pyridine-7-carboxamide.

Period of Performance:

Two months after receipt of approved order.

SCOPE OF WORK

General Requirements:

Independently and not as an agent of the Government, the Contractor shall furnish all the necessary services, qualified personnel, material, equipment, and facilities, not otherwise provided by the Government as needed to perform the Statement of Work below:

Specific Requirements:

- Synthesis of specified two (2) compounds at a 95% purity (confirmed by NMR and LCMS)/
- Shall provide NCATS with 10g, 95% purity of 4-amino-N,N-dicyclopropyl-6-ethyl-1-methyl-1,6-dihydroimidazo[4,5-d]pyrrolo[2,3-b]pyridine-7-carboxamide)

- Shall provide NCATS with 2.5g, 95% purity of N,N-dicyclopropyl-4-((1,5-dimethyl-1H-pyrazol-3-yl)amino)-6-ethyl-1-methyl-1,6-dihydroimidazo[4,5-d]pyrrolo[2,3-b]pyridine-7-carboxamide)
- The purity will be assessed by both NMR and LCMS.

LEVEL OF EFFORT:

Contractor will carry out the synthetic steps that will result in the synthesis of both compounds in this SOW with the appropriate amounts of each as stated.

GOVERNMENT RESPONSIBILITIES

All of the synthetic work will be carried out by contractor in their facilities. Government will pay for completed work.

DELIVERY OR DELIVERABLES

Contractor will deliver both compounds in the appropriate amounts and purity. The work must be completed 2 months after receipt of the approved order. Contractor will provide NCATS biweekly reports of the progress being made (please send to raul.calvo@nih.gov).

REPORTING REQUIREMENTS

Contractor will provide NCATS biweekly reports of the progress being made (please send to raul.calvo@nih.gov).

OTHER CONSIDERATIONS

Travel:

N/A

Key Personnel:

Contractor will provide the personnel responsible for carrying out the synthesis. Raul Calvo (raul.calvo@nih.gov) and Juan Marugan (maruganj@mail.nih.gov) will track progress and provide guidance, as needed, to ensure the compounds are delivered in a timely manner.

Information System Security Plan:

N/A

Data Rights:

The government has sole possession of the compounds generated and any data. Any publications need to be approved by Juan Marugan (maruganj@mail.nih.gov).

Section 508—Electronic and Information Technology Standards:

N/A

Publications and Publicity:

The government has all the rights to any publication and any disclosure of any information generated in the synthesis of both compounds.

Confidentiality of Information:

All the information generated in the synthesis of both compounds is the sole property of the government and so strict confidentiality is expected from the Contractor